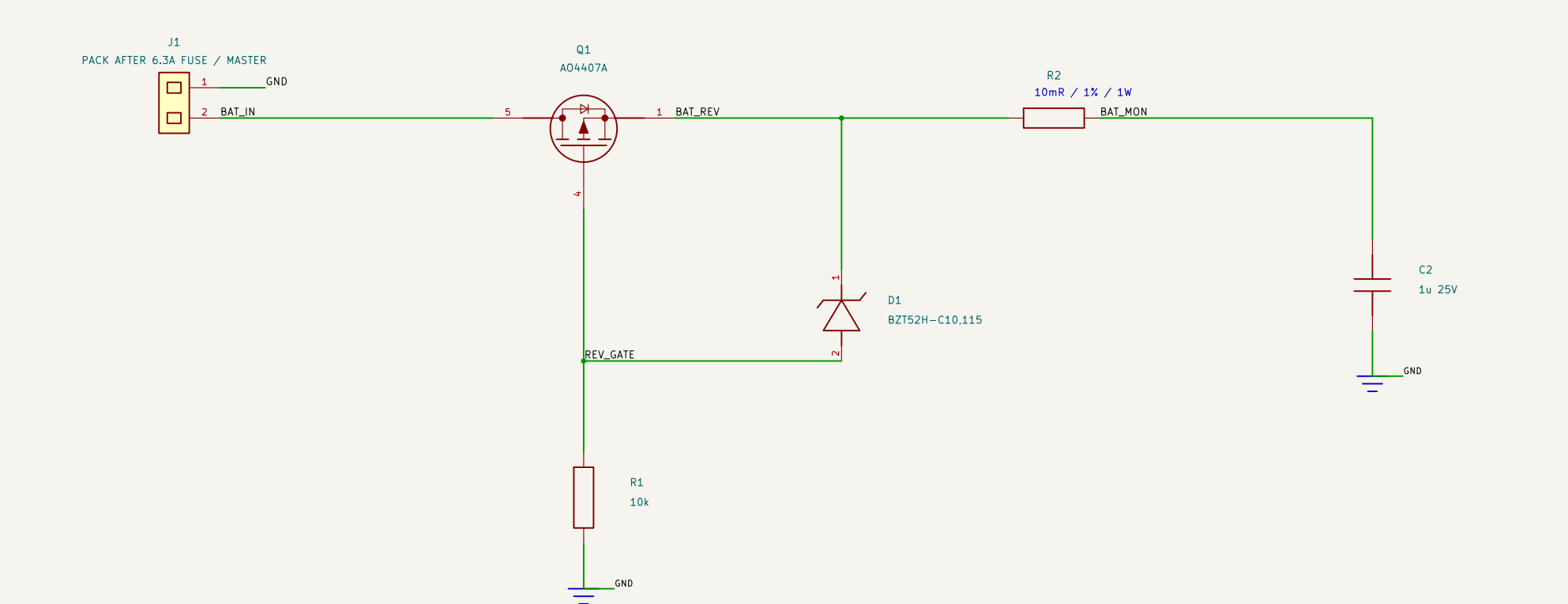


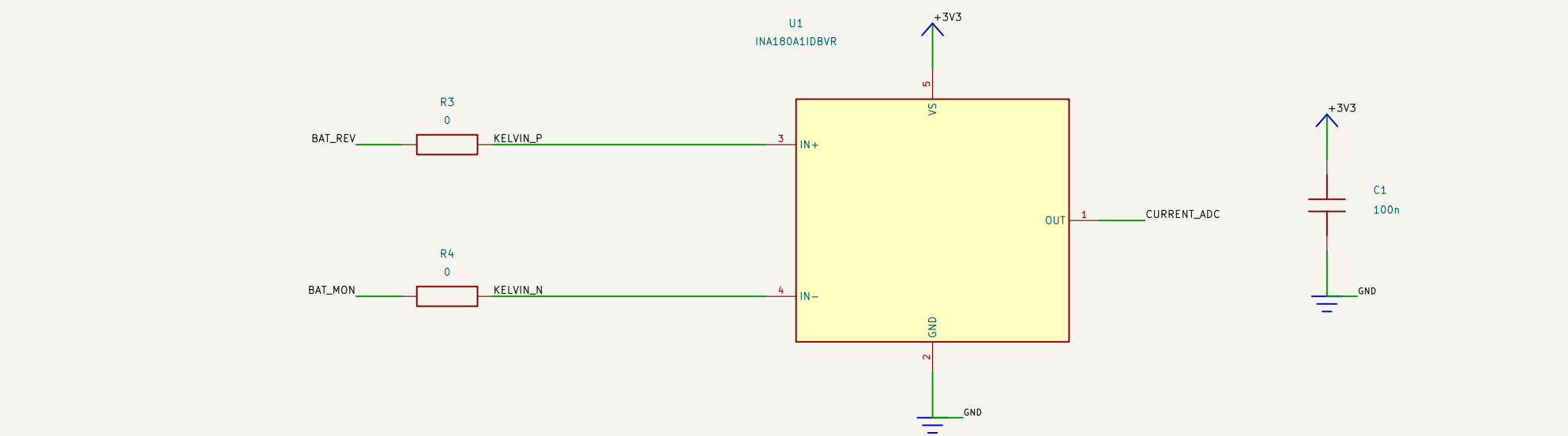
Power flows left to right. Supplies above, returns below. PROTOTYPE / UNVALIDATED.

01 Pack input / reverse protection / shunt



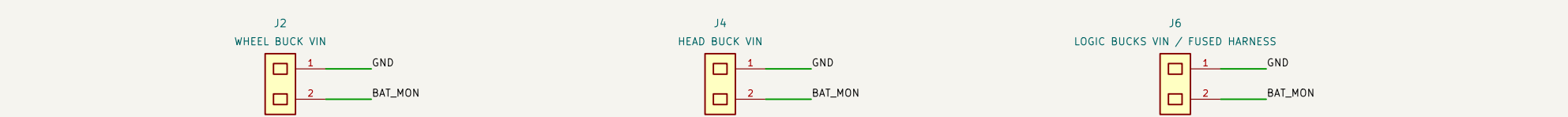
External pack, fuse and master switch precede J1. No charger on this PCB.

02 Kelvin current sense



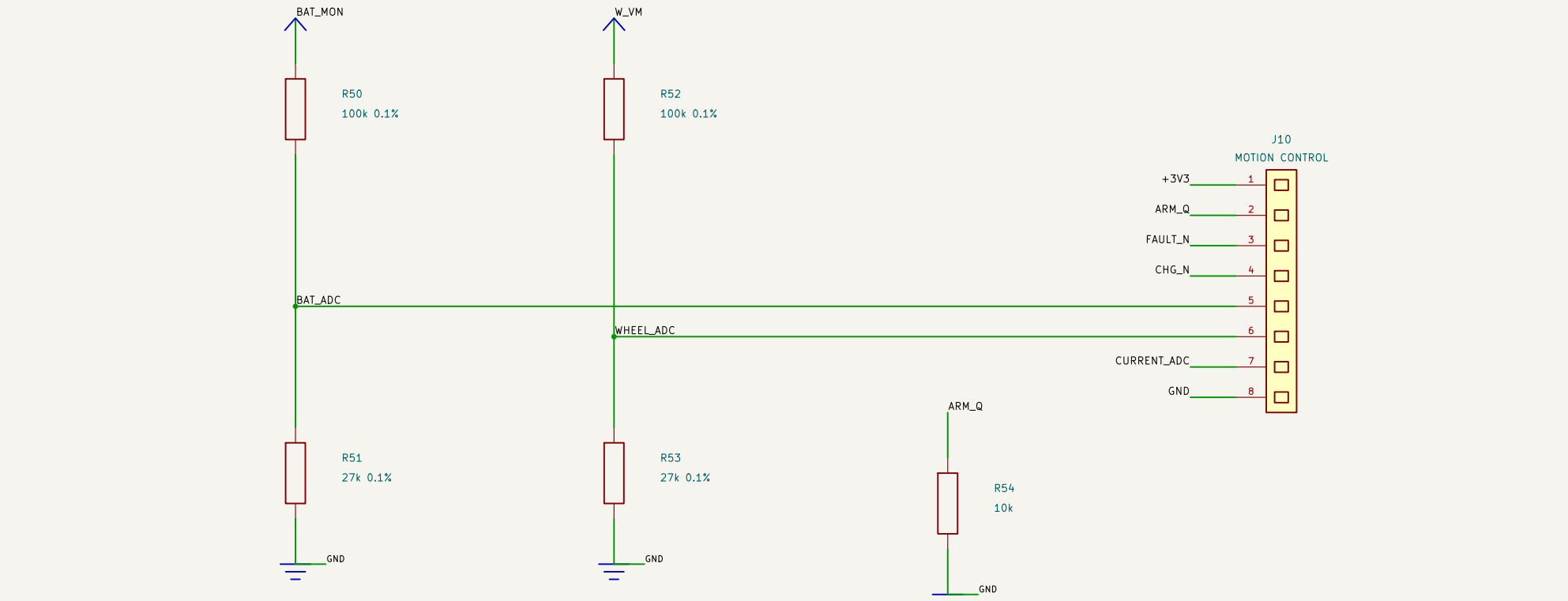
Sense from the shunt pads. Discharge-only nominal transfer: 0.2 V/A.

05 External converter input distribution



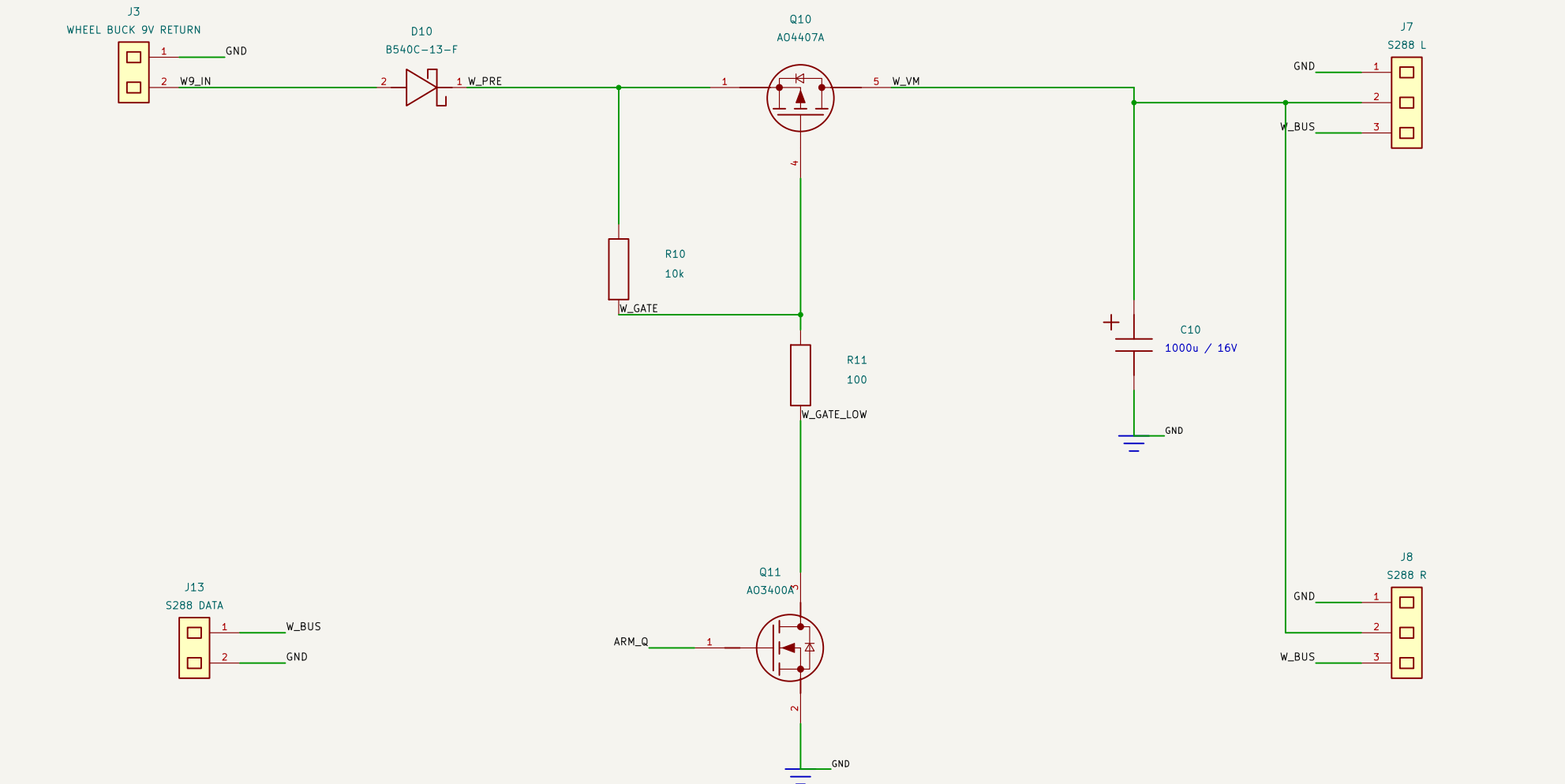
J2 => wheel buck; J4 => head buck; J6 => two independent 5 V converters.

08 Voltage monitor / motion connector



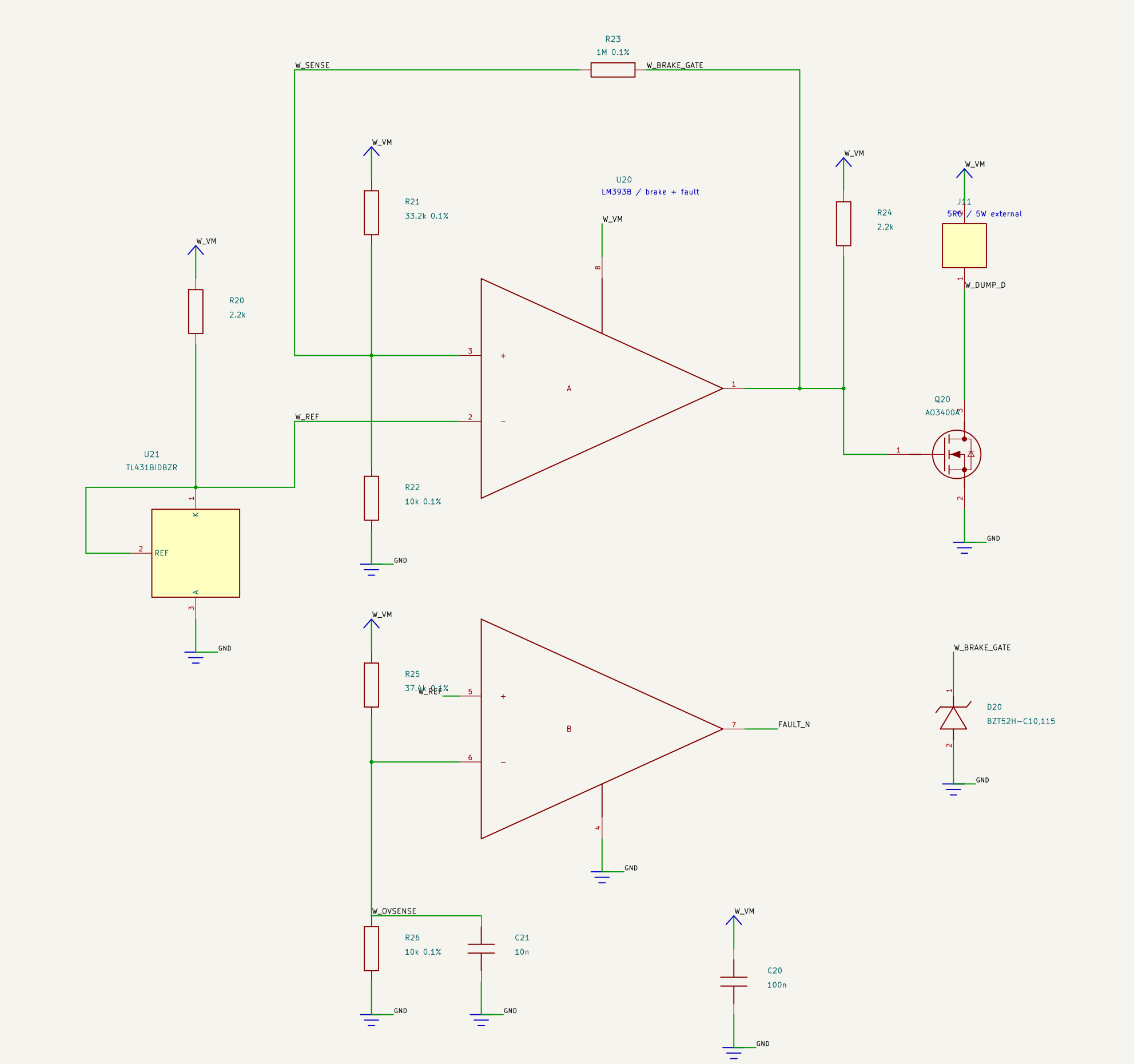
Divider ratio 27 / 127. Current\_ADC comes from block 02.

03 Wheel 9 V conditioning



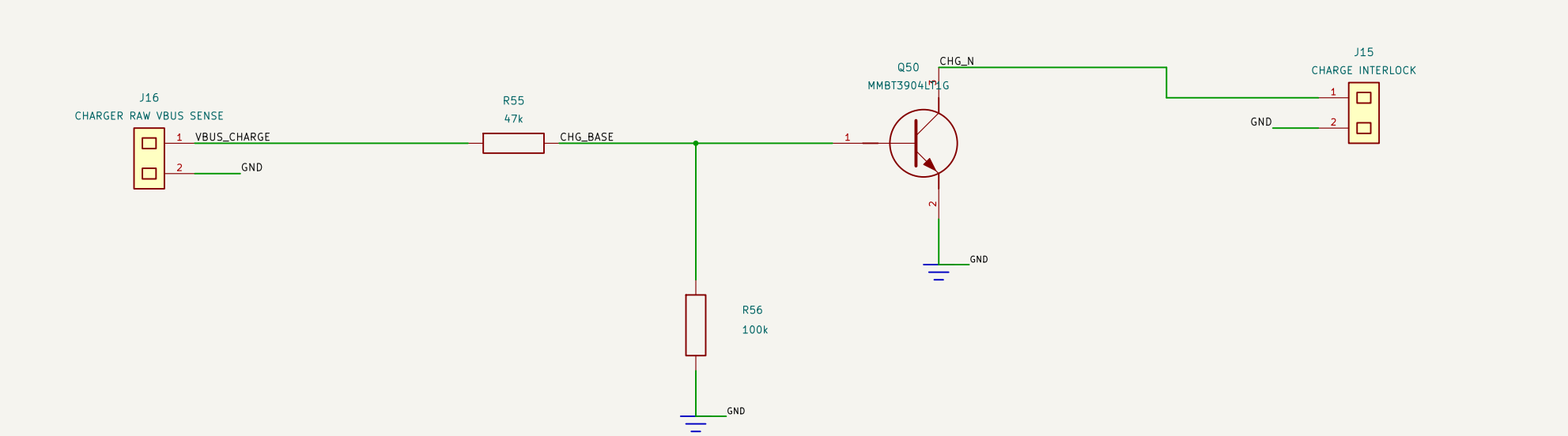
External buck return. Inrush, brake energy and thermal performance: NOT TESTED.

06 Wheel rail / brake and overvoltage



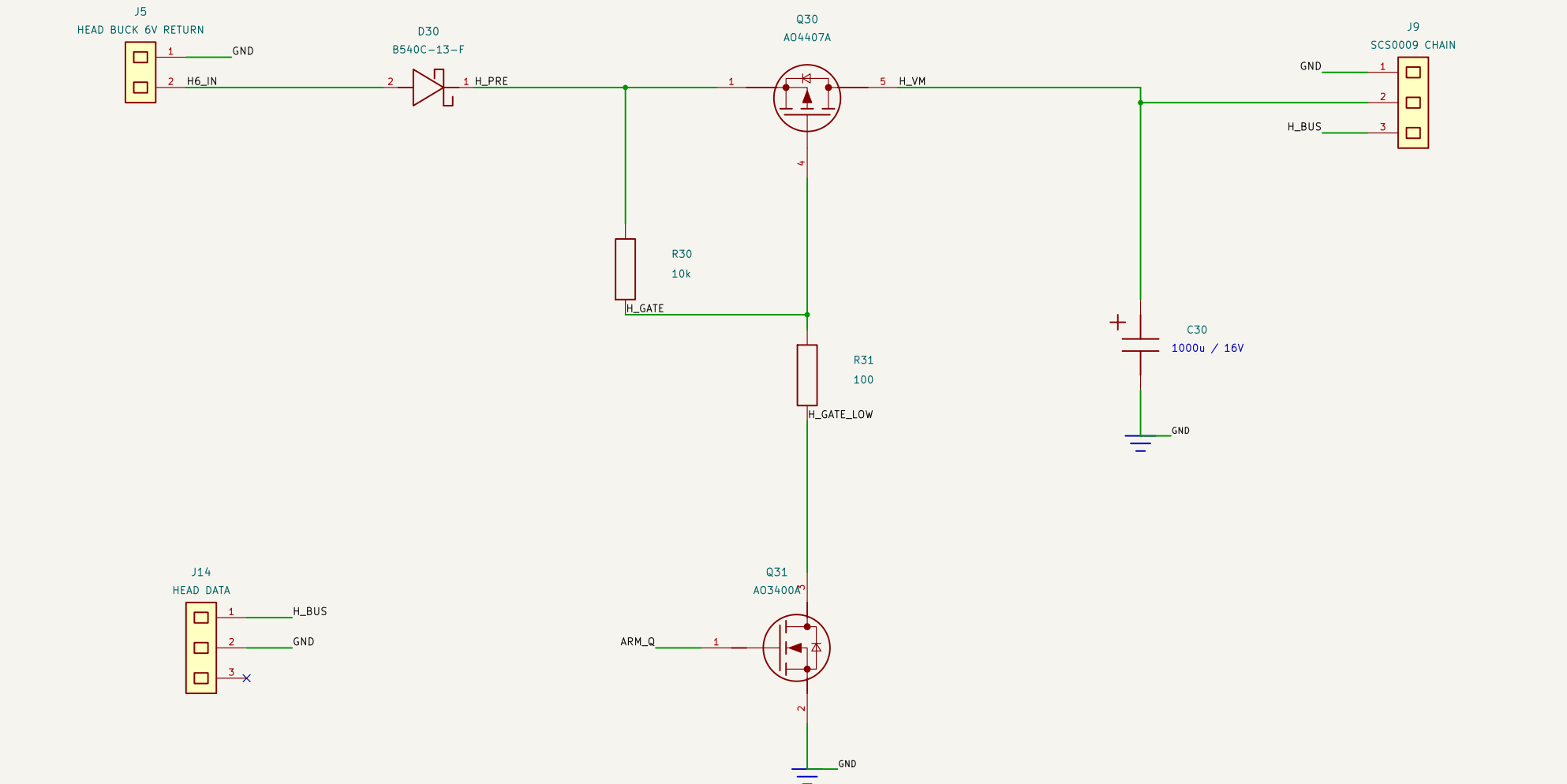
Two comparators share one LM393B. External dump resistor needs pulse qualification.

09 Charge insertion / VBUS sense



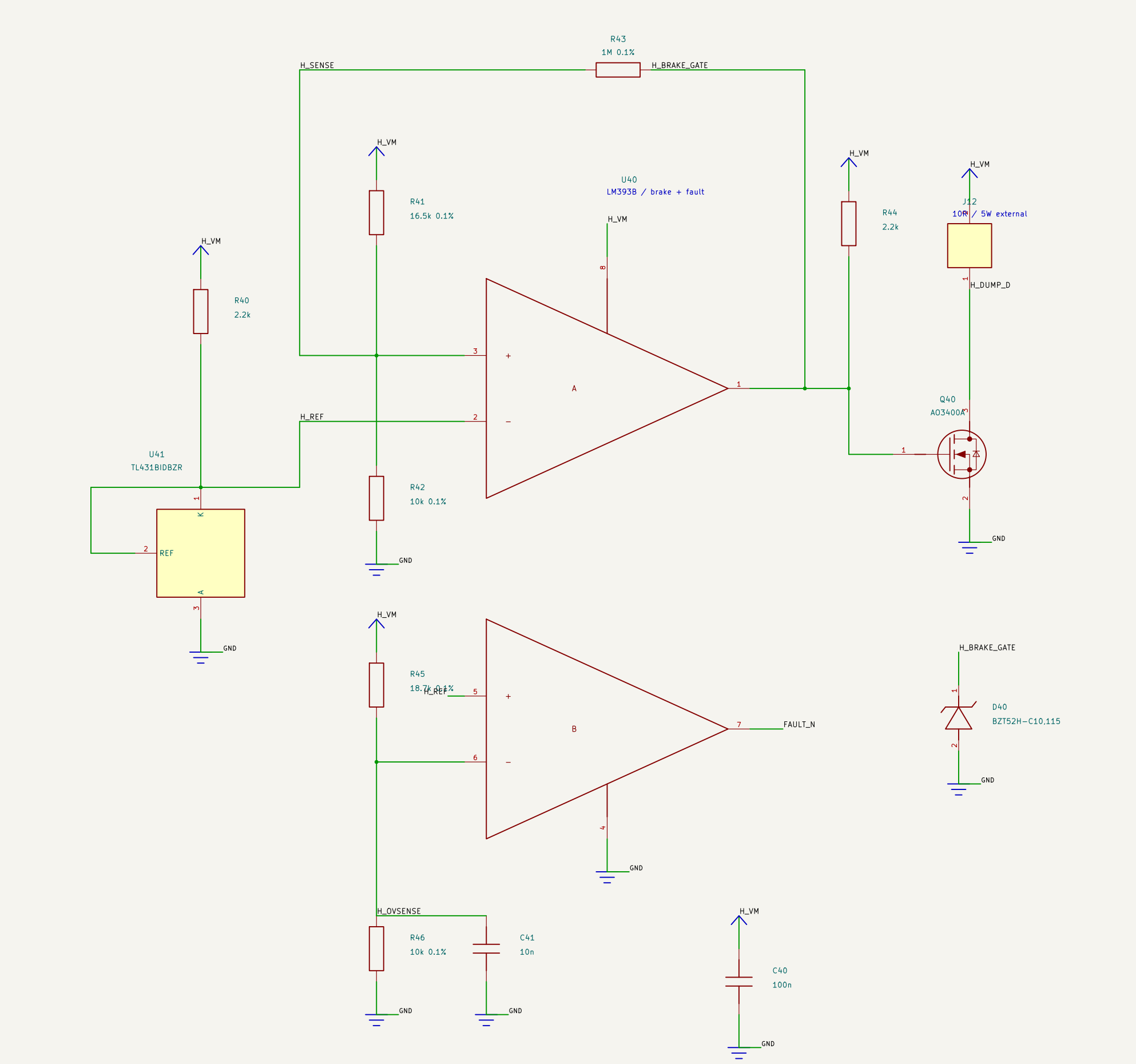
Sense and interlock only: no CC / PD / CC-CV charger implementation.

04 Head 6 V conditioning



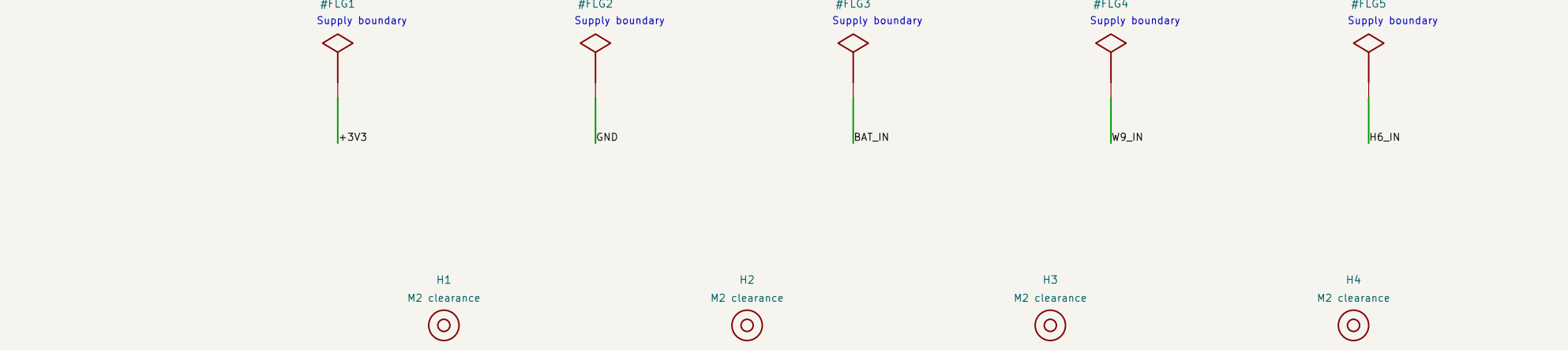
External buck return. Inrush, brake energy and thermal performance: NOT TESTED.

07 Head rail / brake and overvoltage



Two comparators share one LM393B. External dump resistor needs pulse qualification.

10 Supply boundaries / board mounting



80 x 45 mm PCB: populated assembly and thermal performance are not qualified.